arXiv:2312.10253v1 [cs.CL] 15 Dec 2023

Catwalk: A Unified Language Model Evaluation Framework for Many Datasets

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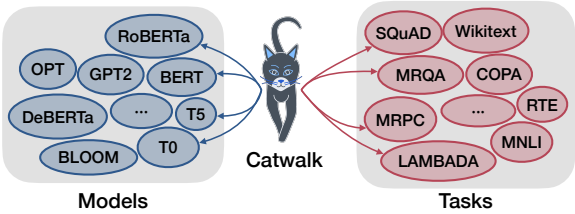
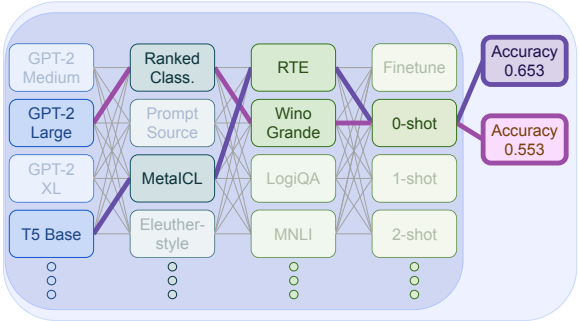
Abstract

The success of large language models has shifted the evaluation paradigms in natural language processing (NLP). The community's interest has drifted towards comparing NLP models across *many* tasks, domains, and datasets, often at an extreme scale. This imposes new engineering challenges: efforts in constructing datasets and models have been fragmented, and their formats and interfaces are incompatible. As a result, it often takes extensive (re)implementation efforts to make fair and controlled comparisons at scale.

Catwalk aims to address these issues. Catwalk provides a unified interface to a broad range of existing NLP datasets and models, ranging from both canonical supervised training and fine-tuning, to more modern paradigms like in-context learning. Its carefully-designed abstractions allow for easy extensions to many others. Catwalk substantially lowers the barriers to conducting controlled experiments at scale. For example, we finetuned and evaluated over 64 models on over 86 datasets with a single command, *without writing any code*. Maintained by the AllenNLP team at the Allen Institute for Artificial Intelligence (AI2), Catwalk is an ongoing open-source effort: https://github.com/allenai/catwalk.



(a) Catwalk provides a unified interfaces to connect models and datasets. This compares n models on m tasks with only $n + m$ custom implementations, substantially reducing the implementation workload compared to current practice where nm implementations are needed.



(b) Illustration of some decisions a user of Catwalk makes to get a result. For example, choosing GPT-2 Large, with Ranked Classification, on WinoGrande gives an accuracy of 0.553, while choosing T5-Base on RTE gives an accuracy of 0.653.

Figure 1: Catwalk and its workflow.

1 Introduction

Large language models (LLMs) have led to a paradigm shift in NLP. Before, a system would be built to tackle a specific task and trained and evaluated on a dedicated dataset. Now, general-purpose pretrained language models are adapted to a broad range of tasks through techniques like fine-tuning and in-context learning. The new paradigm opens research opportunities to investigate models' generalization and robustness across *many* tasks and domains. This imposes new engineering challenges. The fragmented efforts in producing models and datasets have led to countless codebases and data formats that are hardly compatible. Fairly comparing n models on m datasets often require nm custom implementations, substantially raising the barriers to conducting controlled experiments at scale.

Catwalk aims to reduce practitioners' workload in projects involving making fair comparisons among many models and datasets. At a high level, Catwalk can be thought of as a nexus connecting n models and m datasets. Through Catwalk's unified interfaces, training and evaluating all models on all datasets requires $n + m$ customizations instead of mn, many of which have already been implemented. To include a new model or dataset

into Catwalk, one only needs to instantiate a few interfaces, and it can be evaluated on *all* datasets/models that Catwalk supports. Figures 1a and 1b provide illustrative examples. Whenever possible, Catwalk caches and reuses intermediate experimental steps such as loaded datasets/models, model predictions, as well as trained models.[1] This helps minimize redundant machine workloads in, e.g., loading the data and training the models, and better utilize the limited hardware resources. Catwalk's caching mechanism is especially useful in large-scale experiments that involve many models and/or many tasks.

Catwalk is an ongoing effort. We aim to jump-start a virtuous cycle of dataset and model authors adopting support for Catwalk's unified interfaces. To this end, we contribute:

- 6 model abstractions that support a variety of existing models.
- Zero-shot and few-shot evaluations, as well as fine-tuning.
- 9 data formats that cover common use cases.
- Support for 86 standalone datasets as well as curated benchmarks that contain many datasets themselves, for a total of over 800 datasets, along with evaluation metrics.
- A case study that compares 30 widely-used models on 17 datasets.

We expect the above list to keep growing. Catwalk is open-source: `https://github.com/allenai/catwalk`. It is maintained by the AllenNLP team at the Allen Institute for Artificial Intelligence (AI2) and has been a useful tool for their projects, including being the main evaluation framework used in the OLMo (Open Language Model) project.[2] We hope it can benefit the community's research too.

2 Structure and organization

The purpose of Catwalk is to answer the question "How well does model X do on dataset Y?", for as many models and datasets as possible. The possible models include GPT-style decoder-only models (Radford et al., 2018), T5-style encoder/decoder models (Raffel et al., 2019), and BERT-style fine-tuned models (Devlin et al., 2018). Catwalk imposes no constraint on the types of models it can support. The model does not have to be transformer-based or even be a language model at all, though all currently implemented models are transformer-based language models.

To achieve this flexibility, Catwalk transforms instances from a dataset into "formats" that are more suited to model execution. Crucially, these formats are the same across many datasets, so model implementations do not have to concern themselves with the differences between datasets, and can be written in a simple, generic style.

3 Models and Model Wrappers

Catwalk's models combine an underlying model, a model wrapper, and a method to process the outputs from the model. For example, a decoder-only language model can be used to compute the probability of various answer options. This can be done using T0-style human-readable prompts (Sanh et al., 2021), or it can be done using machine-readable prompts in a few-shot setting (Gao et al., 2023). This model could even be fine-tuned and used with a task-specific head (Devlin et al., 2018). These options have different characteristics and produce different results. They are all encapsulated in Catwalk's model wrapper abstraction.

3.1 Model Wrappers

Model wrappers come in two general varieties, *native interfaces* that provide generic support for a wide class of models and *adapter interfaces* that aim to facilitate systematic comparisons with existing model types or evaluation frameworks. We list the different model wrappers below current implemented in Catwalk (with the first three being our native interfaces and the last three being example adapter interfaces; see Table 1).

- **Ranked classification models**. This model wrapper uses decoder-only and encoder/decoder models. It calculates the probability of all answer options, and chooses the most likely answer. It supports zero-shot and few-shot evaluations, and it can be fine-tuned on a dataset, or a list of datasets.
- **BERT-style models**. This model wrapper uses the native API from the Huggingface library (Wolf et al., 2019) to perform multiple-choice, question answering, and classification tasks. This is the way to run encoder-only models like BERT (Devlin et al., 2018), RoBERTa (Liu et al., 2019), or DeBERTa (He

[1] Catwalk's caching mechanism is enabled by AI2-Tango (Groeneveld et al., 2023): `https://github.com/allenai/tango`.

[2] See more details at `https://allenai.org/olmo` as well as `https://github.com/allenai/ai2-olmo-eval` for details about our LLM evaluation efforts.

Model style	Enc-only	Dec-only	Enc/dec	Finetuning	Zero-shot	Few-shot
Ranked classification	✗	✓	✓	✓	✓	✓
Bert-style	✓	✗	✗	✓	✗	✗
Language models	✗	✓	✗	✗	✓	✓
PromptSource	✗	✓	✓	✓	✓	✓
MetaICL	✗	✓	✓	✗	✓	✓
Eleuther-style	✗	✓	✗	✗	✓	✓

Table 1: Model styles and supported features in Catwalk.

et al., 2020). It supports fine-tuning, but no few-shot evaluations.

- **Language models**. This model wrapper is a generic class for decoder-only generation models that supports few- and zero-shot evaluation. This is the class most relevant to recent GPT-style LLM evaluation. In contrast to Eleuther-style models (Gao et al., 2023) *(see below)*, it is designed to support a wider range of generation metrics and token-based scoring strategies beyond those hard-coded in existing toolkits. It also has special features for returning per-instance predictions that allow for more fine-grained analysis and debugging.
- **PromptSource models**. This model wrapper also performs ranked classification, but it uses the human-readable prompts from the PromptSource project (Bach et al., 2022). It supports zero-shot and few-shot evaluations.
- **MetaICL models**. This model wrapper applies the prompt truncation strategy used by Min et al. (2022b) to decoder-only ranked classification. It truncates each of several in-context demonstrations separately and truncates again after concatenation.
- **Eleuther-style models**. This model wrapper uses decoder-only models in the style of Gao et al. (2023). It reproduces those numbers exactly, though it runs a little faster. Like Gao et al. (2023), it supports zero-shot and few-shot evaluations. On many datasets, this is equivalent to the ranked classification models.

3.2 Models

All of Catwalk's existing model implementations load parameters and configurations from Huggingface. They can be configured to load any model from the Huggingface hub, but Catwalk gives a few of them names, tests them regularly, and makes them available on the command line. The following is a list of models that have names in Catwalk and that have been extensively tested:

- BERT {tiny, base, large}, {cased, uncased} (Devlin et al., 2018).
- BLOOM {560M, 1B1, 1B7, 3B, 7B1} (Scao et al., 2022).
- CT0-11b (Scialom et al., 2022).
- DEBERTA {v3-small, v3-base, v3-large, v2-xlarge, v2-xxlarge} (He et al., 2020, 2023).
- DistilBERT-base-cased-distilled-squad (Sanh et al., 2019).
- GPT-J-6B (Wang and Komatsuzaki, 2021).
- GPT-Neo {125M, 1.3B, 2.7B} (Gao et al., 2020).
- GPT-NeoX-20B (Black et al., 2022).
- GPT2 {tiny, regular, medium, large, xl} (Radford et al., 2018).
- mT5 {small, base, large, xl} (Xue et al., 2020).
- OPT {125M, 350M, 1.3B, 2.7B, 6.7B, 13B, 30B, 66B} (Zhang et al., 2022).
- RoBERTa {base, large} (Liu et al., 2019).
- T0 {regular, 3B, original_task_only, single_prompt, p, pp} (Sanh et al., 2021).
- T5 {very_small_random, small, small-lm-adapt, v1_1-small, base, base-lm-adapt, v1_1-base, large, large-lm-adapt, v1_1-large, 3B, xl-lm-adapt, v1_1-xl, 11b, xxl-lm-adapt, v1_1-xxl} (Raffel et al., 2019).
- Pythia {70M,160M,410M,1B,1.4B,2.7B,6.9B,12B} (Biderman et al., 2023)
- Falcon {1B,7B,40B} (Almazrouei et al., 2023)
- Llama(-2) {7B, 13B, 70B} (Touvron et al., 2023a,b)
- MPT {7B, 30B} (MosaicML, 2023)

4 Prompting

Catwalk supports a number of formats for dataset instances. By converting dataset instances into a common format, we allow model implementations to only depend on the format, and not on the vagaries of individual datasets.

Catwalk supports the following formats.

4.1 Huggingface dictionaries

This format makes few guarantees beyond being a Python dictionary. It is the raw format of the data, and rarely useful directly. This is the format that the Huggingface datasets library (Lhoest et al., 2021) produces. It isn't used by any of the models directly, but all the other formats use this as their starting point.

```
{
  "sentence1": "No Weapons of Mass Destruction Found Yet.",
  "sentence2": "Weapons of Mass Destruction Found.",
  "label": 1,
  "idx": 0
}
```

4.2 Huggingface multiple choice

This is a structured format containing a question, a list of answer choices, and optionally the index of the correct answer. This corresponds exactly to the expectation of Huggingface multiple choice models. It is used by the BERT-style models.

```
HFMCInstance(
  id="Mercury_7220990",
  question="Which factor will most likely cause a " +
    "person to develop a fever?",
  answer_choices=[
    "a leg muscle relaxing after exercise",
    "a bacterial population in the bloodstream",
    "several viral particles on the skin",
    "carbohydrates being digested in the stomach"
  ],
  correct_answer_index=1
)
```

4.3 Huggingface question-answering

This is a structured format containing a context, a question, and a list of possible correct answers. This corresponds exactly to the expectation of Huggingface question-answering models. It is used by the BERT-style models.

```
HFQAInstance(
  id="5733be284776f41900661182",
  question="To whom did the Virgin Mary appear in 1858?",
  context="... the Virgin Mary appeared to Saint " +
    "Bernadette Soubirous in 1858 ...",
  answers={
    "text": ["Saint Bernadette Soubirous"],
    "answer_start": [515]})
```

4.4 Huggingface classification

This is a structured format containing a context, as a string or a pair of strings, and an optional label. The possible answer choices are not part of the format but are part of the model that uses these instances. This corresponds exactly to the expectation of Huggingface classification models. It is used by the BERT-style models.

```
HFClassificationInstance(
  text=(
    "No Weapons of Mass Destruction Found Yet.",
    "Weapons of Mass Destruction Found."
  ), label=1)
```

4.5 Eleuther

There are several formats specific to the Eleuther evaluation framework (Gao et al., 2023). They are useful to build other format converters on top of, but are rarely useful on their own. Only the Eleuther-style models use these formats directly.

4.6 T5

This format represents instances as strings starting with the dataset name followed by named fields that make up the instance. It also includes a gold answer that is expected of the model. This corresponds exactly to the way T5 (Raffel et al., 2019) was trained. The format is only supported for a small number of datasets.

```
(
 "rte sentence1: No Weapons of Mass Destruction found yet
    sentence2: Weapons of Mass Destruction Found.",
   "not_entailment"
)
```

4.7 Ranked classification

This format represents instances as a list of options, together with an optional label indicating the correct option. Each option is a text pair, called "context" and "continuation". Models generally compute the probability of the continuation given the context to determine the best answer, though recent work suggests other possibilities (Min et al., 2022a). The format is used by the built-in ranked classification models, and it forms the basis for the PromptSource format.

```
RankClassificationInstance(
  choices=[
    (
      "No Weapons of Mass Destruction Found Yet.\n" +
        "Question: Weapons of Mass Destruction Found. " +
        "True or False?\nAnswer:",
      " True"
    ), (
      "No Weapons of Mass Destruction Found Yet.\n" +
        "Question: Weapons of Mass Destruction Found. " +
        "True or False?\nAnswer:",
      " False")
  ],correct_choice=1)
```

4.8 PromptSource

This format makes the prompts from the PromptSource project (Bach et al., 2022) available in Catwalk. In the PromptSource project, every dataset has multiple prompts. These are reflected in the format by having multiple versions of the instance, one per prompt, in a dictionary. Every possible

version looks exactly like a ranked classification instance, and can be used by model implementations accordingly. The PromptSource model implementation uses this format.

```
{"does the claim": RankClassificationInstance(
    choices=[
      (
        "Does the claim 'Weapons of Mass Destruction " +
          "Found.' follow from the fact that 'No Weapons " +
          "of Mass Destruction Found Yet.'? Please " +
          "answer either yes or no.",
        "yes"
      ), (
        "Does the claim 'Weapons of Mass Destruction '" +
          "Found.' follow from the fact that 'No Weapons " +
          "of Mass Destruction Found Yet.'? Please " +
          "answer either yes or no.", "no"
      )
    ],
    correct_choice=1),
  "imply": RankClassificationInstance(
    choices=[
      (
        "Does 'No Weapons of Mass Destruction Found " +
          "Yet.' imply that 'Weapons of Mass Destruction " +
          "Found.'? Please answer either yes or no.", "yes"
      ), (
        "Does 'No Weapons of Mass Destruction Found " +
          "Yet.' imply that 'Weapons of Mass Destruction " +
          "Found.'? Please answer either yes or no.",
        "no"
      )
    ],correct_choice=1)}
```

4.9 Perplexity

Catwalk also has a general-purpose task format for perplexity evaluation, which was developed as part of the PALOMA benchmark (Magnusson et al., 2023). It follows best practices for perplexity analysis such as avoiding document concatenation (Gao et al., 2020), has advanced batching strategies (e.g., sorting by input length to avoid excessive padding) and supports both non-overlapping and sliding window inference from Press et al. (2020) to handle common issues related to document maximum length truncation.

5 Datasets

The datasets supported by Catwalk can be extended with just a few lines of configs with prompts. Currently Catwalk contains the following datasets:

Perplexity

- WikiText (Merity, 2019)
- LAMBADA (Paperno et al., 2016)

Classification

- CoLA (Warstadt et al., 2019)
- MRPC (Dolan and Brockett, 2005)
- QQP (DataCanary et al., 2017)
- SST-2 (Socher et al., 2013)
- BoolQ (Clark et al., 2019)
- MultiRC (Khashabi et al., 2018)
- WiC (Pilehvar and Camacho-Collados, 2019)
- MC-TACO (Ben Zhou and Roth, 2019)
- PubMedQA (Jin et al., 2019)
- Hendrycks Ethics (Hendrycks et al., 2021a)
- RAFT (Alex et al., 2021)

Open-ended QA

- SQuAD (Rajpurkar et al., 2016)
- SquadShifts (Miller et al., 2020)
- MRQA (Fisch et al., 2019)
- SQuAD2 (Rajpurkar et al., 2018)
- DROP (Dua et al., 2019)
- TriviaQA (Joshi et al., 2017)
- WebQuestions (Berant et al., 2013)
- TruthfulQA (Lin et al., 2021)
- Hendrycks Math (Hendrycks et al., 2021c)
- Arithmetic (Brown et al., 2020)
- Anagrams 1 and 2 (Brown et al., 2020)
- Cycle Letters (Brown et al., 2020)
- Random Insertion (Brown et al., 2020)
- Reversed Words (Brown et al., 2020)
- Natural Questions (Kwiatkowski et al., 2019)

Multiple-choice QA

- PIQA (Bisk et al., 2020)
- COPA (Gordon et al., 2012)
- WSC (Levesque et al., 2011)
- PROST (Aroca-Ouellette et al., 2021)
- SciQ (Welbl et al., 2017)
- QA4MRE (Peñas et al., 2013)
- ARC Easy and Challenge (Clark et al., 2018)
- LogiQA (Liu et al., 2020)
- HellaSwag (Zellers et al., 2019)
- OpenBookQA (Mihaylov et al., 2018)
- RACE (Lai et al., 2017)
- HEAD-QA (Vilares and Gómez-Rodríguez, 2019)
- MathQA (Amini et al., 2019)
- WINOGRANDE (Sakaguchi et al., 2019)
- MuTual (Cui et al., 2020)
- MMLU (Hendrycks et al., 2021b)
- CaseHOLD (Zheng et al., 2021)
- SocialIQA (Sap et al., 2019)
- CSQA (Talmor et al., 2018)

Summarization

- SciTLDR (Cachola et al., 2020)
- XSUM (Narayan et al., 2018)
- EURLEX (Aumiller et al., 2022)

NLI

- RTE from GLUE (Wang et al., 2018)
- RTE from SuperGLUE (Wang et al., 2019)
- MNLI (Williams et al., 2018)
- QNLI (Wang et al., 2018)
- WNLI (Wang et al., 2018)

- CB (de Marneffe et al., 2019)
- Adverserial NLI (Nie et al., 2020)

Mixtures of tasks

- P3 (Sanh et al., 2022)
- MetaICL (Min et al., 2022b)

Note that some of these datasets are collections that show up in Catwalk as multiple subsets. For example, there are actually 11 RAFT datasets, each taking a different angle on language model evaluation. In total, this adds up to over 800 datasets.

Adding new datasets to Catwalk is easy. There are classes for including datasets from the Huggingface hub, and format conversion functions can often be written in a single line of code.

6 Evaluation

In Catwalk, the responsibility for calculating metrics lies with the model implementations. Model implementations can calculate any metric that makes sense in the context of the model. However, for a given dataset we often want the same metrics independent of the model, so we can compare models against each other. To facilitate this, Catwalk allows one to attach metrics to datasets, which act as a suggestion to the models for which metrics to compute. All implemented model styles except the Eleuther style compute the suggested metrics.

By default, Catwalk computes the following metrics and can be easily extended to include others.

- For multiple-choice, classification, and entailment datasets: Accuracy and relative improvement over the random baseline
- For question-answering datasets: SQuAD metric as defined by Rajpurkar et al. (2016).
- For language modeling datasets: Perplexity per word, perplexity per byte, and entropy.

7 Case Studies

We include some examples of the kinds of analyses Catwalk enables and in the end describe some ongoing work. Table 3 in the Appendix gives an example of the dense matrix of models and datasets that Catwalk can generate. There are many research questions that can be posed and answered with a matrix of results like this:

What strategy for adapting pretrained models to downstream tasks performs best overall? For instance we can compare finetuning an encoder-only model against zero-shot evaluation on autoregressive models. We find that, macro averaged over 17 datasets, the best finetuned models had 62% greater relative improvement in accuracy over a random baseline than the zero-shot models , although the largest zero-shot model is an order of magnitude larger than the largest finetuned model.

Which datasets are easy/hard? *All* zero-shot and finetuned models perform well on SciQ, with an average of more than 200% relative accuracy improvement compared to the random-guess baseline. Among the datasets in our experiments, QNLI is the most challenging one for zero-shot ranked classification models, with all models barely outperforming random guesses. Unsurprisingly, all finetuned models substantially outperform the chance baseline on all datasets, and in all cases they outperform zero-shot models by a wide margin.

What impact does model type or size have across datasets? Among zero-shot models, we see that, while T5 variants are the only ones that significantly outperform the chance baseline on MNLI, QQP, and RTE, they *underperform* others and even the random baseline on CoLA and COPA. This could be attributed to the different choices T5 models make in terms of learning objectives, pretraining data, etc., which requires further investigation. Also, in both the zero-shot ranked classification and finetuning settings, larger model sizes, in general, lead to better performance *within the same model family*, with few exceptions.

Is the ordering of models by performance consistent across datasets? Table 2 in the Appendix shows that every dataset puts the models into roughly the same order by accuracy, but correlations are often weak. The largest correlations are observed between the ARC Easy, ARC Challenge, and OpenbookQA datasets. All three of these are multiple-choice datasets targeting science questions. On the other end of the spectrum, LogiQA and SciQ correlate the least with other datasets, with coefficients below 0.5. This might be explained by LogiQA being the most difficult, and SciQ being the easiest dataset among the 17 that were part of this study. In one case the models cannot get much signal out of the dataset, and in the other they saturate. A thorough investigation in this style would have to include many more models and prompting methods.

These analyses only scratch the surface of the questions that Catwalk can help answer. There are

many open questions that bear investigation:

- How well does perplexity predict performance on other datasets?
- Do the zero-shot results hold in few-shot settings? How does the number of few-shot examples affect the results?
- How important is the format of the prompt?
- Some large language models make intermediate checkpoints available for analysis. At what point in model training do abilities emerge?

Ongoing work As mentioned at the outset, Catwalk is currently be used as the main evaluation framework for the OLMo project at AI2. As such, it has many features that allow it to be a replacement for other evaluation frameworks, such as Eleuther Harness (Gao et al., 2023). For more details about Catwalk for LLM evaluation, see `https://github.com/allenai/ai2-olmo-eval`.

8 Related Work

Holistic Evaluation of Language Models (HELM; Liang et al., 2022) also aims to increase the density of coverage between models and evaluations and take a complementary path to ours. HELM describes itself as a top-down approach; they design a taxonomy of scenarios and metrics and use them to curate a specific subset of combinations. Catwalk is bottom-up, and aims to lower the engineering barriers for model and task creators to integrate their work into a shared benchmark. For example, HELM chooses to only adapt models via a standardized 5-shot prompting method, while Catwalk explicitly incorporates different varieties of prompting and even fine-tuning.

Our work builds on the EleutherAI LM Evaluation Harness (Gao et al., 2023). It similarly addresses the task of building a unified evaluation framework and focuses on decoder-only models and zero-/few-shot evaluations. By contrast, Catwalk also includes encoder-only and encoder/decoder models and supports fine-tuning.

9 Conclusion

We presented Catwalk, a system to evaluate many models on many datasets. Catwalk's APIs eliminate redundant implementation efforts and unify incompatible approaches to language model evaluation. The approach enables experimentation across datasets and language models at an unprecedented scale. While a large number of models and datasets is already integrated, the Allen Institute for Artificial Intelligence continues to provide hands-on engineering help to grow this list, and invites further contributions as an open-source effort.

Limitations

Language model evaluation can only be as good as the datasets that are used. Catwalk takes no position on this topic, and instead makes as many datasets as possible available for study. As shown in Table 2, the choice of dataset can have a big influence on which model appears "best". Similarly, prompted models are often sensitive to the prompt, and fine-tuned models are sensitive to their training regime. We hope that Catwalk will enable the kind of research that allows us to take strong, supported positions on these topics in the future.

Ethics Statement

We believe the impact of this framework to be largely beneficial, in that it encourages the community to adopt reproducible research practices. A unified interface for evaluation allows for easy comparison across various papers, without the need for reimplementation and rerunning experiments that may take up significant time and compute resources.

We also note that the datasets present in our framework may contain biases and artifacts of their own, and a model performing well within our framework on these datasets should not be considered as having "solved" the underlying task.

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A Appendix

	ARC Challenge	ARC Easy	CoLA	COPA	HEAD-QA EN	HellaSwag	LogiQA	MNLI	MRPC	OpenBookQA	PIQA	QNLI	QQP	RTE	SciQ	SST	WINOGRANDE
ARC Challenge	1.000	0.932	0.606	0.811	0.800	0.826	0.587	0.716	0.642	0.939	0.614	0.852	0.696	0.796	0.543	0.577	0.747
ARC Easy	0.932	1.000	0.650	0.896	0.860	0.862	0.492	0.649	0.652	0.914	0.713	0.744	0.585	0.693	0.603	0.502	0.805
CoLA	0.606	0.650	1.000	0.616	0.440	0.381	0.618	0.622	0.864	0.600	0.244	0.525	0.433	0.503	0.086	0.590	0.475
COPA	0.811	0.896	0.616	1.000	0.727	0.822	0.442	0.482	0.552	0.765	0.641	0.617	0.475	0.535	0.454	0.429	0.784
HEAD-QA EN	0.800	0.860	0.440	0.727	1.000	0.778	0.295	0.438	0.501	0.804	0.811	0.531	0.336	0.525	0.691	0.249	0.751
HellaSwag	0.826	0.862	0.381	0.822	0.778	1.000	0.238	0.532	0.422	0.772	0.825	0.666	0.499	0.608	0.679	0.398	0.892
LogiQA	0.587	0.492	0.618	0.442	0.295	0.238	1.000	0.570	0.634	0.515	0.085	0.482	0.506	0.516	−0.040	0.483	0.204
MNLI	0.716	0.649	0.622	0.482	0.438	0.532	0.570	1.000	0.675	0.773	0.311	0.835	0.822	0.788	0.222	0.712	0.420
MRPC	0.642	0.652	0.864	0.552	0.501	0.422	0.634	0.675	1.000	0.648	0.240	0.586	0.540	0.556	0.137	0.685	0.455
OpenBookQA	0.939	0.914	0.600	0.765	0.804	0.772	0.515	0.773	0.648	1.000	0.581	0.874	0.769	0.828	0.520	0.608	0.726
PIQA	0.614	0.713	0.244	0.641	0.811	0.825	0.085	0.311	0.240	0.581	1.000	0.417	0.235	0.308	0.641	0.160	0.774
QNLI	0.852	0.744	0.525	0.617	0.531	0.666	0.482	0.835	0.586	0.874	0.417	1.000	0.897	0.828	0.381	0.750	0.562
QQP	0.696	0.585	0.433	0.475	0.336	0.499	0.506	0.822	0.540	0.769	0.235	0.897	1.000	0.855	0.251	0.801	0.413
RTE	0.796	0.693	0.503	0.535	0.525	0.608	0.516	0.788	0.556	0.828	0.308	0.828	0.855	1.000	0.403	0.719	0.523
SciQ	0.543	0.603	0.086	0.454	0.691	0.679	−0.040	0.222	0.137	0.520	0.641	0.381	0.251	0.403	1.000	0.089	0.583
SST	0.577	0.502	0.590	0.429	0.249	0.398	0.483	0.712	0.685	0.608	0.160	0.750	0.801	0.719	0.089	1.000	0.311
WINOGRANDE	0.747	0.805	0.475	0.784	0.751	0.892	0.204	0.420	0.455	0.726	0.774	0.562	0.413	0.523	0.583	0.311	1.000

Table 2: Spearman correlations between models when ranked in order of performance on 17 common datasets.

	Models	ARC Challenge	ARC Easy	CoLA	COPA	HEAD-QA	HellaSwag	LogiQA	MNLI	MRPC	OpenBookQA	PIQA	QNLI	QQP	RTE	SciQ	SST	WINOGRANDE
Zero-shot Ranked Classification	BLOOM 560M	−4.1	60.4	9.9	24.0	47.0	42.7	6.3	6.4	36.8	23.2	30.6	−0.5	−24.8	5.4	190.0	30.7	2.4
	BLOOM 1B1	3.1	79.3	38.3	32.0	51.5	63.7	3.2	6.6	36.8	24.8	36.1	−1.0	−26.4	5.4	198.0	23.6	9.4
	BLOOM 1B7	5.5	90.1	38.3	38.0	64.5	84.0	4.5	6.5	36.8	28.8	38.1	−1.0	−26.4	6.9	198.8	7.3	14.0
	BLOOM 3B	15.0	109.6	37.1	40.0	67.0	108.8	10.0	6.7	36.8	40.8	41.7	−1.1	−26.4	12.6	223.2	2.1	17.0
	BLOOM 7B1	27.6	127.6	32.3	32.0	77.6	136.2	4.5	7.0	36.8	58.4	47.2	2.4	−26.3	8.3	231.2	−1.6	28.7
	GPT-J 6B	43.0	140.1	35.4	54.0	89.6	163.6	17.4	12.2	36.8	68.0	52.2	2.9	−23.5	9.0	238.4	−1.4	28.2
	GPT-2	−8.2	52.7	33.1	34.0	30.2	23.7	0.2	1.2	13.2	8.8	23.6	0.3	−25.4	6.1	167.6	8.0	3.2
	GPT-2 Medium	−1.4	70.7	38.3	30.0	39.7	57.2	8.8	5.5	36.8	20.8	35.1	−1.1	−26.3	5.4	168.4	11.9	6.2
	GPT-2 Large	−2.4	83.2	37.9	36.0	41.7	77.4	13.1	7.7	30.4	18.4	39.3	−1.3	−25.6	5.4	182.0	−1.1	10.7
	GPT-2 XL	11.6	103.7	38.3	40.0	47.9	99.2	5.7	9.6	26.5	39.2	41.8	2.9	−25.6	4.7	193.2	−1.8	16.7
	OPT 125M	−10.9	54.5	38.3	26.0	30.4	25.2	14.9	3.5	36.8	8.0	23.8	−1.2	−26.5	−0.4	186.8	6.2	0.6
	OPT 350M	−6.8	55.9	38.3	30.0	34.4	44.1	13.1	3.3	36.8	14.4	29.9	−0.9	−26.4	4.0	178.8	23.4	5.4
	OPT 1.3B	9.6	96.3	21.6	46.0	48.4	113.7	0.8	7.5	32.8	32.0	45.3	2.8	−21.7	3.2	198.0	63.5	19.2
	OPT 2.7B	18.8	117.2	−30.2	46.0	56.8	138.8	5.7	6.5	35.8	53.6	49.6	2.4	−13.4	9.0	206.0	3.4	22.2
	OPT 6.7B	33.8	136.4	−5.3	58.0	61.4	168.4	11.8	−1.6	24.5	56.8	54.2	1.8	−21.6	10.5	226.4	52.8	30.5
	T5 Small	−14.3	17.8	31.0	−4.0	16.3	6.2	10.0	7.8	37.3	11.2	12.1	−0.6	−20.9	−6.1	143.2	41.7	−0.7
	T5 Base	−11.3	18.4	−37.3	−4.0	30.0	22.3	12.4	77.3	−36.8	27.2	11.3	0.9	26.4	30.7	172.4	14.7	0.2
	T5 Large	0.3	22.2	−38.3	2.0	34.8	55.8	1.4	5.4	−36.8	30.4	11.3	1.1	26.4	10.5	224.0	0.5	13.0
	T5 3B	9.9	74.6	−21.2	6.0	43.1	91.2	4.5	61.8	35.3	28.0	35.9	7.4	35.3	40.8	255.2	77.5	1.3
	T5 11B	27.6	72.1	−38.3	18.0	47.5	109.0	10.0	43.6	−33.8	31.2	43.7	14.0	26.2	29.2	186.8	75.5	5.1
Finetuned MLMs	BERT Base Cased	33.1	116.2	68.4	38.0	54.3	53.8	26.0	152.0	72.5	107.2	33.2	80.4	81.4	36.5	184.8	86.2	3.9
	BERT Large Cased	38.9	135.5	70.3	36.0	57.0	82.2	29.6	159.0	73.5	132.8	40.4	83.3	82.0	50.2	198.0	87.8	28.5
	BERT Base Uncased	34.8	122.9	67.8	40.0	57.0	60.8	33.3	152.9	73.0	132.0	30.7	82.6	82.0	37.9	182.0	87.4	4.3
	BERT Large Uncased	38.6	136.5	68.9	48.0	54.3	91.8	37.0	159.0	74.5	134.4	18.3	83.2	82.5	51.6	194.0	89.0	3.9
	DeBERTa-v3 Small	73.4	163.8	73.3	58.0	68.3	198.2	26.6	163.4	82.4	140.0	58.3	83.7	83.8	54.5	223.6	89.0	32.8
	DeBERTa-v3 Base	118.4	200.0	75.1	78.0	87.1	252.8	51.2	170.2	78.9	184.0	64.0	88.1	84.7	71.8	238.0	92.0	65.1
	DeBERTa-v3 Large	178.8	246.5	77.2	96.0	107.1	276.1	72.7	172.9	82.4	232.8	76.0	90.3	85.6	83.4	254.8	92.0	75.7
	DeBERTa-v2 XLarge	34.1	204.4	71.4	48.0	67.4	269.4	1.4	171.8	77.9	200.0	60.0	86.8	83.9	74.0	233.6	93.8	73.3
	RoBERTa Base	34.8	115.8	67.2	44.0	55.0	151.1	25.3	162.3	79.4	113.6	39.0	83.4	83.0	59.6	158.4	90.1	31.2
	RoBERTa Large	75.8	158.4	74.1	80.0	43.7	233.5	31.5	170.6	80.4	109.6	25.4	87.9	83.2	76.9	196.0	93.3	54.1

Table 3: Improvement relative in percentage compared to the random baseline on 17 common tasks